

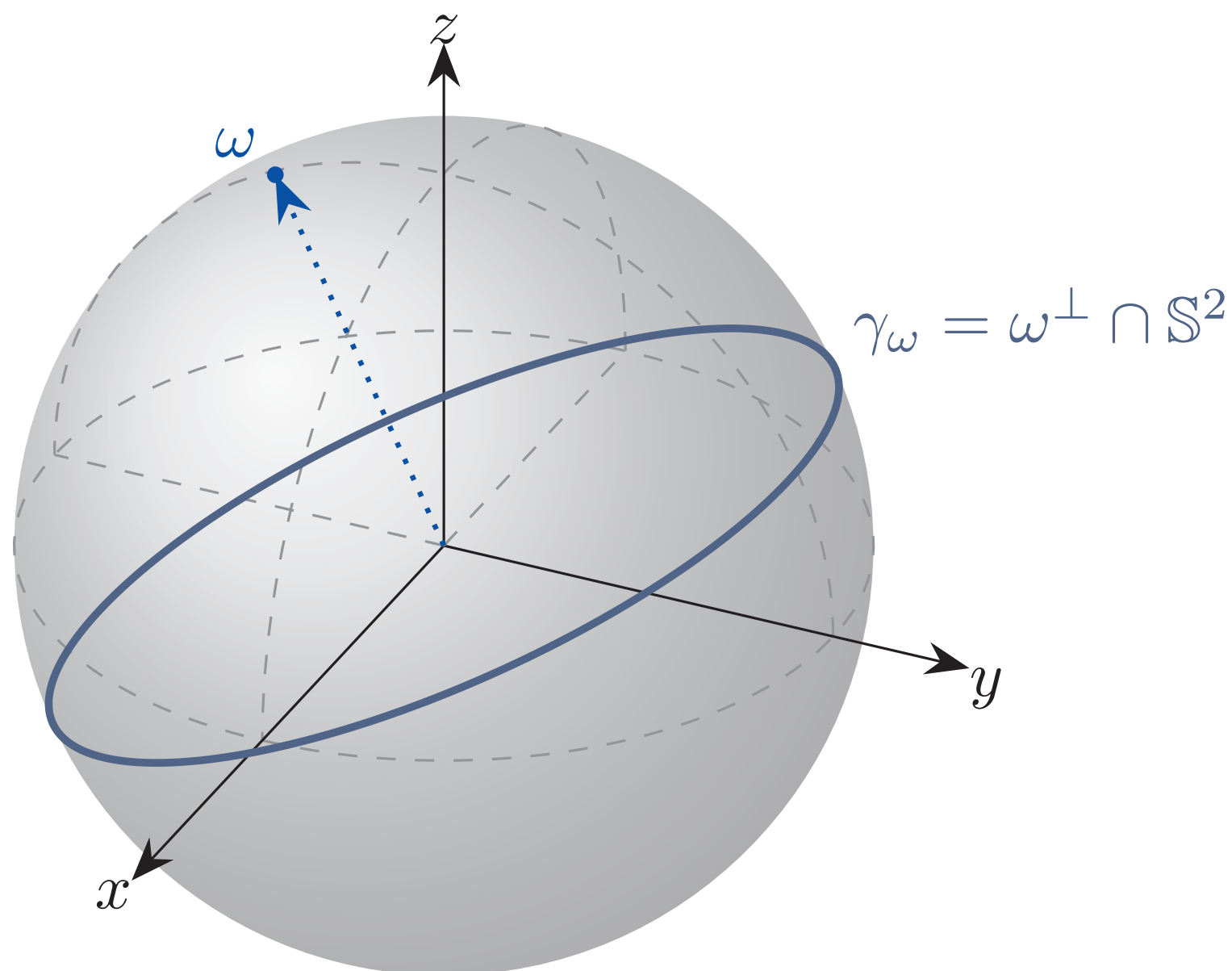
1. THE FUNK TRANSFORM

In 1911 Paul Funk introduced the following transform over the sphere. For $\omega \in \mathbb{S}^2$ and $f \in C^\infty(\mathbb{S}^2)$ let

$$\gamma_\omega = \{\sigma \in \mathbb{S}^2 : \omega \cdot \sigma = 0\},$$

be the great circle from ω . Then the *Funk transform* is

$$(\mathcal{F}f)(\omega) = \int_{\gamma_\omega} f(\sigma) ds(\sigma) = \int_0^{2\pi} f(\gamma_\omega(t)) dt.$$



Funk's results:

- $\mathcal{F}$ annihilates odd functions;
- $\mathcal{F}$ is invertible on even functions, and for every $Y \in \mathcal{H}_{2\ell}$,
$$\mathcal{F}[Y] = \lambda_{2\ell} Y, \quad \boxed{\lambda_{2\ell} = 2\pi P_{2\ell}(0)}.$$

4. LEGENDRE POLYNOMIALS

For each degree ℓ , the spherical harmonic space $\mathcal{H}_\ell$ has basis

$$\{Y_{\ell m} : -\ell \leq m \leq \ell\}.$$

In spherical coordinates $\sigma = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$, and when $m = 0$, spherical harmonic functions take the form

$$Y_{\ell 0}(\theta, \phi) = \sqrt{\frac{2\ell+1}{4\pi}} P_\ell(\cos \theta),$$

where P_ℓ is the Legendre polynomial of degree ℓ .

The Legendre polynomials are given by the formula

$$P_\ell(x) = \frac{1}{2^\ell} \sum_{k=0}^{\lfloor \ell/2 \rfloor} (-1)^k \binom{\ell}{k} \binom{2\ell-2k}{\ell} x^{\ell-2k}.$$

For example, $P_0(t) = 1$, $P_1(t) = t$ and $P_2(t) = \frac{1}{2}(3t^2 - 1)$.

Their parity is determined by the degree:

$$P_\ell(-t) = (-1)^\ell P_\ell(t).$$

Consequently, $P'_{2k}(0) = 0$, while $P'_{2k+1}(0) = (-1)^k \frac{(2k+2)!}{2^{2k+1}k!(k+1)!}$.

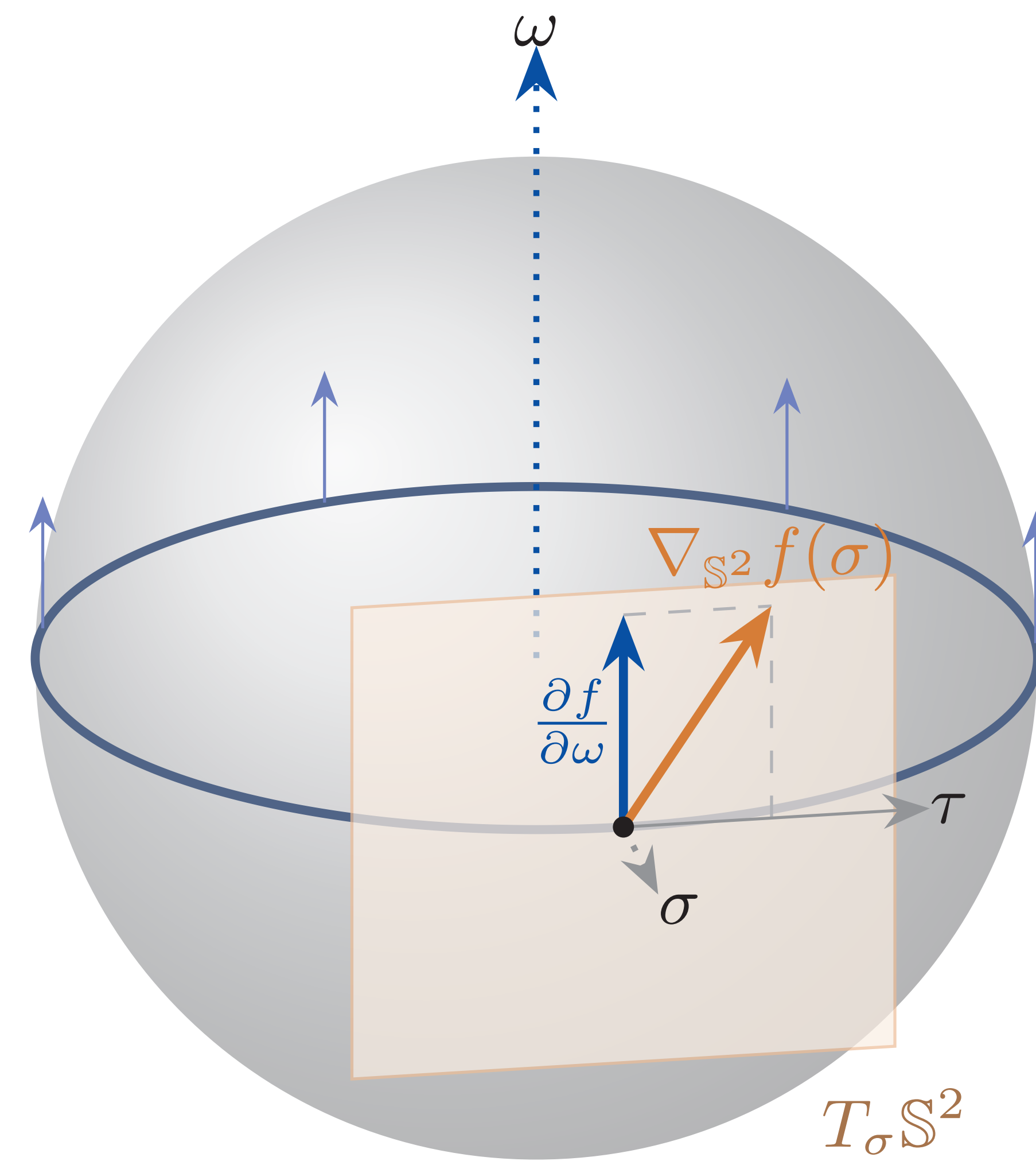
On the equator $\theta = \pi/2$, so $\cos \theta = 0$. Thus $P'_\ell(0)$ describes how $Y_{\ell 0}$ changes in the direction normal to the equator, which is exactly the quantity appearing in the eigenvalues of the G -transform.

2. THE G -TRANSFORM: A NORMAL DERIVATIVE ALONG GREAT CIRCLES

Definition. For $f \in C^1(\mathbb{S}^2)$ and $\omega \in \mathbb{S}^2$,

$$\mathcal{G}[f](\omega) = \int_{\omega \cdot \sigma = 0} \langle \nabla_{\mathbb{S}^2} f(\sigma), \omega \rangle ds(\sigma) = \int_{\gamma_\omega} \frac{\partial f}{\partial \omega}(\sigma) ds(\sigma).$$

This is the Funk transform of the normal derivative.



Symmetries of the $\mathcal{G}$ -transform. 1) Replacing ω by $-\omega$ fixes the circle while reverses the normal. So $\mathcal{G}[f](-\omega) = -\mathcal{G}[f](\omega)$. 2) Rotating the picture rotates the integral, so

$$\mathcal{G}[R \cdot f] = R \cdot \mathcal{G}[f] \quad (R \in \text{SO}(3)), \quad (R \cdot f)(\sigma) = f(R^{-1}\sigma).$$

Hence, every output of G is odd, and G commutes with rotations.

5. MAIN RESULT

Diagonalisation theorem.

For every $Y \in \mathcal{H}_\ell$,

$$\mathcal{G}[Y] = \lambda_\ell Y, \quad \boxed{\lambda_\ell = 2\pi P'_\ell(0)}.$$

Since $\mathcal{G}$ acts by a scalar on $\mathcal{H}_\ell$, it suffices to only compute

$$\mathcal{G}[Y_{\ell 0}](e_3) = \lambda_\ell Y_{\ell 0}(e_3),$$

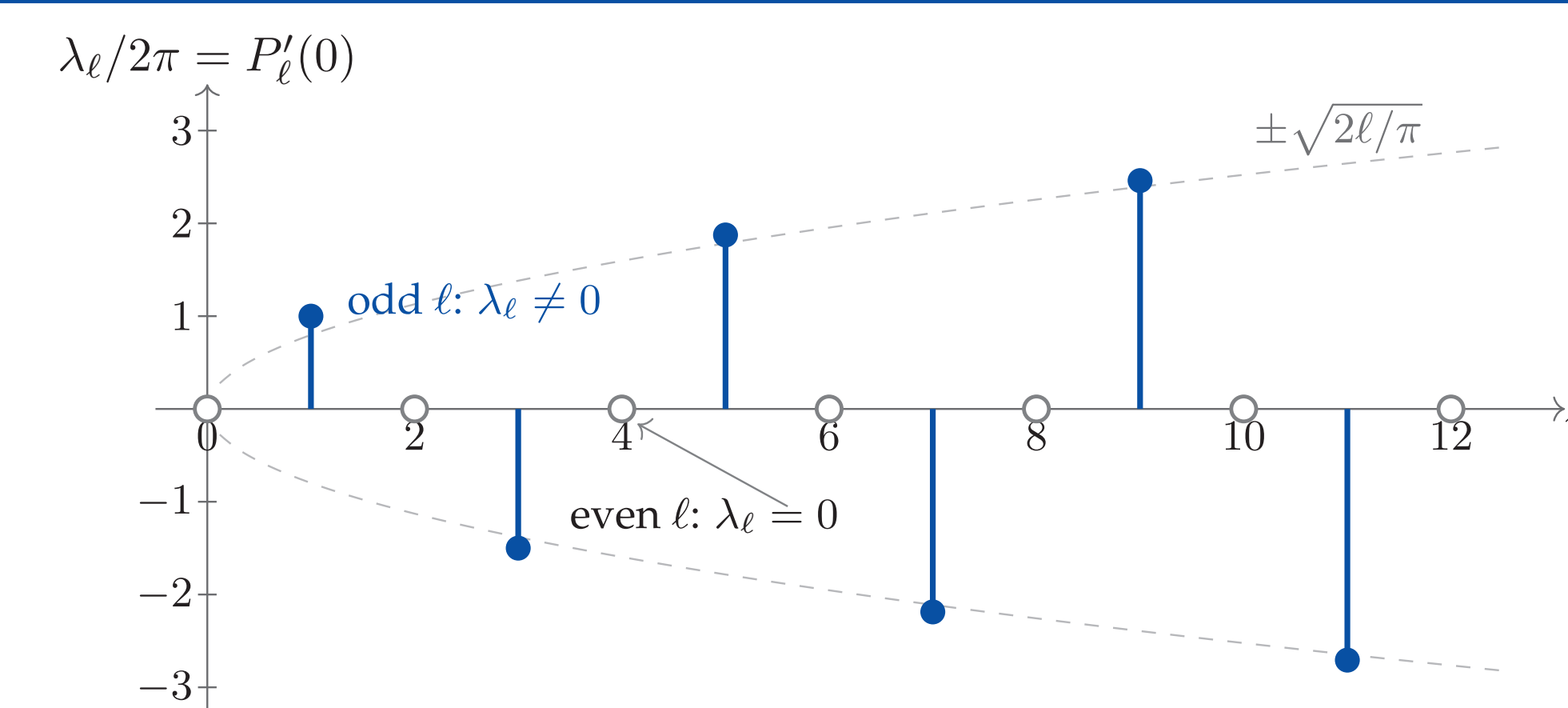
where e_3 is the north pole. So for $\ell = 2k + 1$,

$$\lambda_\ell = 2\pi P'_\ell(e_3) = 2\pi(-1)^k \frac{(2k+2)!}{2^{2k+1}k!(k+1)!} \neq 0.$$

Thus $\mathcal{G}$ is invertible on the odd subspace. In particular,

$\ker \mathcal{G} = C^\infty_{\text{even}}(\mathbb{S}^2)$, $\mathcal{G} : C^\infty_{\text{odd}}(\mathbb{S}^2) \rightarrow C^\infty_{\text{odd}}(\mathbb{S}^2)$ is invertible.

6. THE SPECTRUM AT A GLANCE



The eigenvalues are zero on even degrees and alternating on odd degrees, with $|\lambda_\ell| \sim 2\sqrt{2\ell/\pi}$.

	even modes	odd modes
Funk transform $\mathcal{F}$	isomorphism	kernel
G -transform $\mathcal{G}$	kernel	isomorphism

Complementary: $C^\infty(\mathbb{S}^2) = \ker \mathcal{F} \oplus \ker \mathcal{G}$.

3. SPHERICAL HARMONICS

The spherical Laplacian decomposes

$$L^2(\mathbb{S}^2) = \bigoplus_{\ell \geq 0} \mathcal{H}_\ell, \quad \Delta_{\mathbb{S}^2} Y = -\ell(\ell+1)Y$$

for $Y \in \mathcal{H}_\ell$, with $\dim \mathcal{H}_\ell = 2\ell + 1$.

Each $\mathcal{H}_\ell$ is an *irreducible* representation of $\text{SO}(3)$. Since $\mathcal{F}$, $\mathcal{G}$ and Δ are $\text{SO}(3)$ -equivariant, Schur's lemma says $\mathcal{F}|_{\mathcal{H}_\ell} = \lambda'_\ell I$ and $\mathcal{G}|_{\mathcal{H}_\ell} = \lambda_\ell I$. Hence, we can invert the transform by

$$f = \sum_{\ell \geq 0} f_\ell \xrightarrow{\mathcal{G}} \sum_{\ell \geq 0} \lambda_\ell f_\ell$$

$$f = \sum_{\ell \geq 0} f_\ell \xrightarrow{\mathcal{F}} \sum_{\ell \geq 0} \lambda'_\ell f_\ell$$

7. REGULARITY AND INVERSION

For odd ℓ , $|\lambda_\ell| \asymp \ell^{1/2} \asymp (1 + \ell(\ell+1))^{1/4}$.

Thus, on the odd subspace, $\mathcal{G}$ has Sobolov scaling of an operator of order $1/2$.

Consequently, for every $s \in \mathbb{R}$,

$$\mathcal{G} : H^{s+1/2}_{\text{odd}}(\mathbb{S}^2) \rightarrow H^s_{\text{odd}}(\mathbb{S}^2)$$

is an isomorphism, with two-sided bounds

$$\|\mathcal{G}f\|_{H^s} \asymp \|f\|_{H^{s+1/2}}.$$

Inversion. Given $M \in H^s_{\text{odd}}(\mathbb{S}^2)$, the equation

$$\mathcal{G}N = M$$

has a *unique* odd solution $N \in H^{s+1/2}_{\text{odd}}(\mathbb{S}^2)$, and

$$\|N\|_{H^{s+1/2}} \lesssim \|M\|_{H^s}.$$

Therefore, $\mathcal{G}N = M$ gains half a derivative over the data and the even part of N is never determined, since it lies in $\ker \mathcal{G}$.

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